

HTTP Request packet

```
▼ Frame 30078: Packet, 484 bytes on wire (3872 bits), 484 bytes captured (3872 bits) on interface \Device\NPF_{D8FCD502-3369-466A-9F5B-2698866835B4}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{D8FCD502-3369-466A-9F5B-2698866835B4})
  Encapsulation type: Ethernet (1)
  Arrival Time: Jul 18, 2026 23:08:32.073853800 Bangladesh Standard Time
  UTC Arrival Time: Jul 18, 2026 17:08:32.073853800 UTC
  Epoch Arrival Time: 1784394512.073853800
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 99.600 microseconds]
  [Time since reference or first frame: 1 minute, 52.248135800 seconds]
  Frame Number: 30078
  Frame Length: 484 bytes (3872 bits)
  Capture Length: 484 bytes (3872 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http]
  Character encoding: ASCII (0)
  [Coloring Rule Name: HTTP]
  [Coloring Rule String: http || tcp.port == 80 || http2]
```

Frame: This is the frame info, the physical/capture layer. Wireshark shows this packet is 484 bytes, came in through the Ethernet interface, and tags it as HTTP since it can tell the stack is Ethernet → IP → TCP → HTTP.

```
✓ Ethernet II, Src: HonHaiPrecis_f1:11:87 (7c:66:ef:f1:11:87), Dst: TPLink_cf:8d:40 (c0:3a:55:cf:8d:40)
  ▼ Destination: TPLink_cf:8d:40 (c0:3a:55:cf:8d:40)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  > Source: HonHaiPrecis_f1:11:87 (7c:66:ef:f1:11:87)
    Type: IPv4 (0x0800)
    [Stream index: 0]
```

Ethernet II: This is the Data Link layer. It just shows two MAC addresses, my PC's network card (Intel) as the source and my router (TP-Link) as the destination. MAC addresses only handle the trip from my PC to the router, not the whole way to the server.

```
✓ Internet Protocol Version 4, Src: 192.168.0.116, Dst: 172.67.132.115
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 470
    Identification: 0xe0a8 (57512)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x26a6 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.116
    Destination Address: 172.67.132.115
    [Stream index: 13]
```

IP: This is the Network layer. My pc 's local IP (192.168.0.116) is the source, and the website's IP (172.67.132.115) is the destination. It also shows the TTL is 128 and that TCP is the next layer up.

```
▼ Transmission Control Protocol, Src Port: 59448, Dst Port: 80, Seq: 1, Ack: 1, Len: 430
  Source Port: 59448
  Destination Port: 80
  [Stream index: 44]
  [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 430]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 1857182346
  [Next Sequence Number: 431      (relative sequence number)]
  Acknowledgment Number: 1      (relative ack number)
  Acknowledgment number (raw): 1819357325
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 255
  [Calculated window size: 65280]
  [Window size scaling factor: 256]
  Checksum: 0x23f9 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (430 bytes)
```

TCP: This is the Transport layer. My pc uses port 59448, talking to the server's port 80. The PSH and ACK flags mean this packet is pushing new data (my request) while also confirming earlier data was received.

```
✓ Hypertext Transfer Protocol
  > GET / HTTP/1.1\r\n
  Host: httpforever.com\r\n
  Connection: keep-alive\r\n
  Upgrade-Insecure-Requests: 1\r\n
  User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36\r\n
  Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
  Accept-Encoding: gzip, deflate\r\n
  Accept-Language: en-US,en;q=0.9\r\n
  \r\n
  [Response in frame: 30120]
  [Full request URI: http://httpforever.com/]
```

HTTP: This is the actual request, the Application layer. It's a GET request for the homepage of httpforever.com. The User-Agent line shows I was using Chrome on Windows, and the Accept line shows what file types my browser is willing to receive back.

HTTP Response packet

```
▼ Frame 30120: Packet, 59 bytes on wire (472 bits), 59 bytes captured (472 bits) on interface \Device\NPF_{D8FCD502-3369-466A-9F5B-2698866B35B4}, id 0
  Section number: 1
  > Interface id: 0 (\Device\NPF_{D8FCD502-3369-466A-9F5B-2698866B35B4})
    Encapsulation type: Ethernet (1)
    Arrival Time: Jul 18, 2026 23:08:32.546108900 Bangladesh Standard Time
    UTC Arrival Time: Jul 18, 2026 17:08:32.546108900 UTC
    Epoch Arrival Time: 1784394512.546108900
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 1.500 microseconds]
    [Time delta from previous displayed frame: 472.255100 milliseconds]
    [Time since reference or first frame: 1 minute, 52.720390900 seconds]
    Frame Number: 30120
    Frame Length: 59 bytes (472 bits)
    Capture Length: 59 bytes (472 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

Frame: Frame info for the response. It's 59 bytes, arrived about 472.255100 milliseconds after the request, and Wireshark tags it as HTTP too. This time with "data-text-lines" added, since the response has readable text in it.

```
▼ Ethernet II, Src: TPLink_cf:8d:40 (c0:3a:55:cf:8d:40), Dst: HonHaiPrecis_f1:11:87 (7c:66:ef:f1:11:87)
  ▼ Destination: HonHaiPrecis_f1:11:87 (7c:66:ef:f1:11:87)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  ▼ Source: TPLink_cf:8d:40 (c0:3a:55:cf:8d:40)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

Ethernet II: Ethernet layer for the response, and the MAC addresses are flipped now. My router (TP-Link) is the source, and my pc (Intel) is the destination, since the reply is coming back in from the internet through the router.

```

▼ Internet Protocol Version 4, Src: 172.67.132.115, Dst: 192.168.0.116
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x04 (DSCP: LE, ECN: Not-ECT)
    Total Length: 45
    Identification: 0x41c1 (16833)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 59
    Protocol: TCP (6)
    Header Checksum: 0x0c33 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 172.67.132.115
    Destination Address: 192.168.0.116
    [Stream index: 13]

```

IP: IP layer for the response; addresses are flipped here too. The server (172.67.132.115) is now the source, sending to my pc (192.168.0.116). The TTL is 59 this time, lower than the request, since it started from a different default on the server's side.

```

▼ Transmission Control Protocol, Src Port: 80, Dst Port: 59448, Seq: 5204, Ack: 431, Len: 5
  Source Port: 80
  Destination Port: 59448
  [Stream index: 44]
  [Stream Packet Number: 11]
  > [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 5]
  Sequence Number: 5204 (relative sequence number)
  Sequence Number (raw): 1819362528
  [Next Sequence Number: 5209 (relative sequence number)]
  Acknowledgment Number: 431 (relative ack number)
  Acknowledgment number (raw): 1857182776
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 16
  [Calculated window size: 131072]
  [Window size scaling factor: 8192]
  Checksum: 0x1505 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (5 bytes)
  TCP segment data (5 bytes)

```

TCP: TCP layer for the response. Ports are flipped from the server's port 80 back to my port 59448. PSH and ACK are set again, since the server is pushing its response data and confirming what it got from me.

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Content-Type: text/xml; charset="utf-8"\r\n
    Connection: close\r\n
  ▼ Content-Length: 2616\r\n
    [Content length: 2616]
    Server: TP-LINK/TP-LINK UPnP/1.1 MiniUPnPd/1.8\r\n
    Ext:\r\n
    \r\n
    [Request in frame: 418025]
    [Time since request: 4.049400 milliseconds]
    [Request URI: /vvied/rootDesc.xml]
    [Full request URI: http://192.168.0.1:1900/vvied/rootDesc.xml]
    File Data: 2616 bytes
```

HTTP: This is the actual response. Status code 200 OK means it worked. Here, the content type is text/xml.

Summary

Basically, each layer just wraps the one before it, like putting a letter inside an envelope after envelope. My browser writes the HTTP request. TCP wraps that with port numbers and a sequence number, so the server knows where to send the reply and how to keep the data in order. IP wraps that again with the source and destination IP addresses, so it actually knows where to send it across the internet. Ethernet wraps it one last time with MAC addresses, but that part only gets it from my pc to my router, the first stop along the way. The response comes back through the same layers in reverse: Ethernet gets it from the router to my pc, IP shows it's really coming from the server, TCP matches it up using the port and sequence numbers, and HTTP is what finally shows the 200 OK and the actual page content inside.